ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S [ANSWER KEY]

Unit 6.01: The Laplace Transform, Step Forcing & Convolution

Q1. Find mathcal L{3 t + cos{leftt right}} .

[ ] A: frac{2 s^{3} + 3 s^{2} + s + 3}{s^{4
[ ] B: frac{- s^{3} - 3 s^{2} - 3}{s^{4} +
[ ] C: frac{s^{3} + 3 s^{2} + 3}{s^{3} + s}
[x] D: frac{s^{3} + 3 s^{2} + 3}{s^{4} + s^

Solution: Use the table and linearity: mathcal L{3 t + cos{leftt right}}=frac{s^{3} + 3 s^{2} + 3}{s^{4} + s^{2}} .

Q2. Using the definition, mathcal L{e^{2 t}}=int_0^infty e^{-s-2t},dt equals for s>2 :

[ ] A: frac{s}{s - 2}
[x] B: frac{1}{s - 2}
[ ] C: frac{1}{s + 2}
[ ] D: - frac{1}{s - 2}

Solution: Evaluate the improper integral; the boundary term at infinity vanishes for s>2 , leaving frac{1}{s - 2} .

Q3. If mathcal L{y}=Ys , y0=-3 and y'0=-1 , find mathcal L{y''} .

[ ] A: Y s^{2} + s + 3
[ ] B: Y s^{2} - 3 s - 1
[x] C: Y s^{2} + 3 s + 1
[ ] D: Y s + 4

Solution: mathcal L{y''}=s^2Ys-sy0-y'0 = Y s^{2} + 3 s + 1 .

Q4. Find mathcal L{t^{4}} .

[ ] A: frac{24}{s^{4}}
[x] B: frac{24}{s^{5}}
[ ] C: frac{4}{s^{5}}
[ ] D: frac{24}{s^{6}}

Solution: Use the table and linearity: mathcal L{t^{4}}=frac{24}{s^{5}} .

Q5. Using the definition, mathcal L{2 t}=int_0^infty 2te^{-st},dt equals for s>0 :

[ ] A: frac{2}{s}
[ ] B: frac{2}{s^{3}}
[x] C: frac{2}{s^{2}}
[ ] D: - frac{2}{s^{2}}

Solution: Evaluate the improper integral; the boundary term at infinity vanishes for s>0 , leaving frac{2}{s^{2}} .

Q6. The second-order linear homogeneous ODE f''x - 53f'x + 696fx = 0 has characteristic roots lambda_1, lambda_2 . What is the trace sum lambda_1 + lambda_2 ?

[x] A: 53
[ ] B: 56
[ ] C: 77
[ ] D: 28

Solution: From lambda^2 - 53lambda + 696 = 0 , the sum of roots is 53 .

Q7. If mathcal L{y}=Ys and y0=1 , find mathcal L{y'} .

[x] A: Y s - 1

[ ] B: Y s + 1

[ ] C: Y - 1

[ ] D: Y s

Solution: mathcal L{y'}=sYs-y0 = Y s - 1 .

Q8. Find mathcal L{t^{3}} .

[x] A: frac{6}{s^{4}}

[ ] B: frac{6}{s^{3}}

[ ] C: frac{3}{s^{4}}

[ ] D: frac{6}{s^{5}}

Solution: Use the table and linearity: mathcal L{t^{3}}=frac{6}{s^{4}} .